

FORM TP 2010176



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MAY/JUNE 2010

CARIBBEAN EXAMINATIONS COUNCIL
ADVANCED PROFICIENCY EXAMINATION

ENVIRONMENTAL SCIENCE

UNIT 2: AGRICULTURE, ENERGY AND ENVIRONMENTAL POLLUTION

PAPER 02

2 hours 30 minutes

21 MAY 2010 (p.m.)

READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. **DO NOT** open this examination paper until instructed to do so.
2. This paper consists of **NINE** questions, **THREE** from each Module.
3. Candidates **MUST** answer **SIX** questions, **TWO** from **EACH** Module.
4. Write your answers in the answer booklet provided.
5. The use of a silent, non-programmable, scientific calculator is permitted.

MODULE 1

Answer ANY TWO questions.

1. (a) A farmer in Trinidad who was experiencing some problems with pests on his farm sought the advice of the agriculture extension officer. The officer recommended that the farmer use the biological pest control method. Study Figure 1, which illustrates the outcome of the application of this method of pest control, and answer the questions that follow.

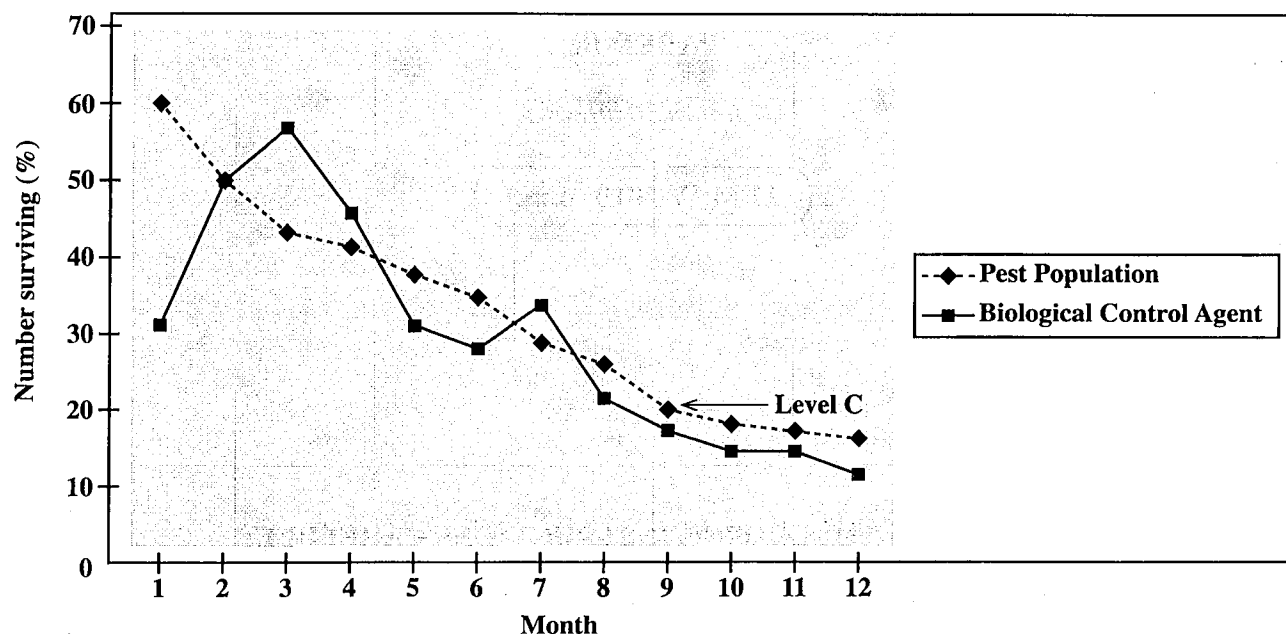


Figure 1. Biological pest control on a farm

- (i) Describe the trend shown in Figure 1. [4 marks]
- (ii) State THREE pieces of information that are vital when employing this method of pest control. [3 marks]
- (iii) Suggest THREE reasons why this method of pest control was recommended. [6 marks]
- (iv) Why would the agriculture officer advise the farmer to try always to maintain the biological control agent at the level C shown in Figure 1? [3 marks]
- (b) The farmer on a nearby farm decided to continue to employ chemical control methods. Why do you think that the farmer made this decision in spite of his knowledge of biological pest control? Suggest TWO reasons for your answer. [4 marks]

Total 20 marks

2. (a) Distinguish between 'genetic engineering' and 'plant and animal breeding'. [3 marks]
- (b) Study Figure 2, which shows the change in the agricultural labour force and the average size of farms in a Caribbean country over the period 1998 to 2006, and answer the questions that follow.

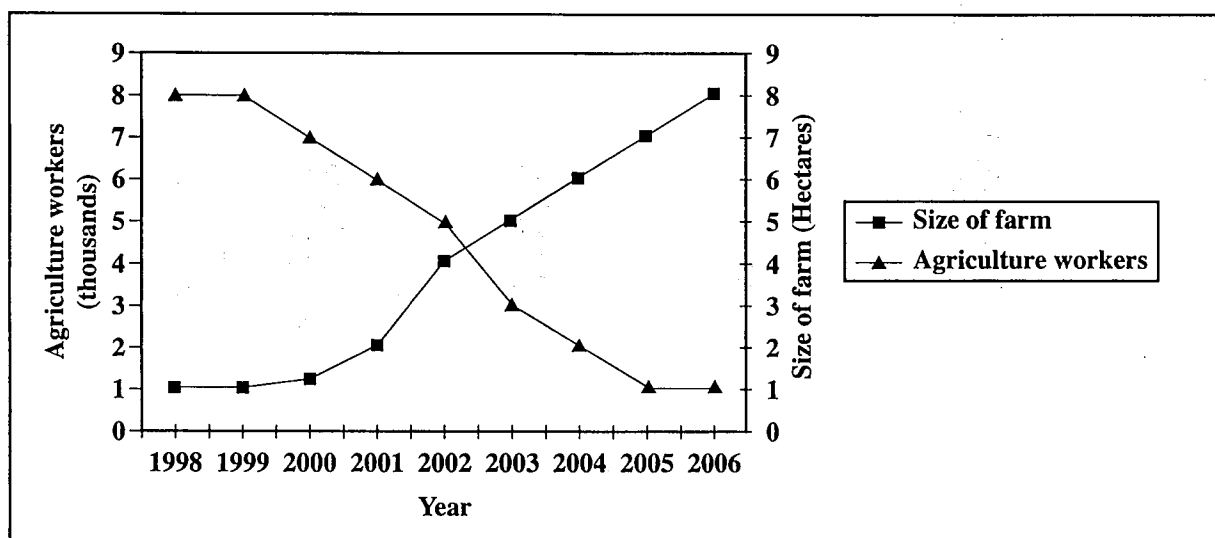


Figure 2. Change in the agricultural labour force and size of farms

- (i) Describe the trend shown in Figure 2. [4 marks]
- (ii) In which year was the average size of farms 4 hectares? [1 mark]
- (iii) Suggest THREE reasons for the trend observed in Figure 2. [6 marks]
- (c) Many Caribbean countries are recommending the practice of sustainable agriculture in an effort to increase their agricultural output and so guarantee continued food security in the region.
- Suggest TWO reasons for the recommendation above. [6 marks]

Total 20 marks

3. (a) Study Figure 3, which shows the causes of soil degradation at two farms, Farm A and Farm B, and answer the questions that follow.

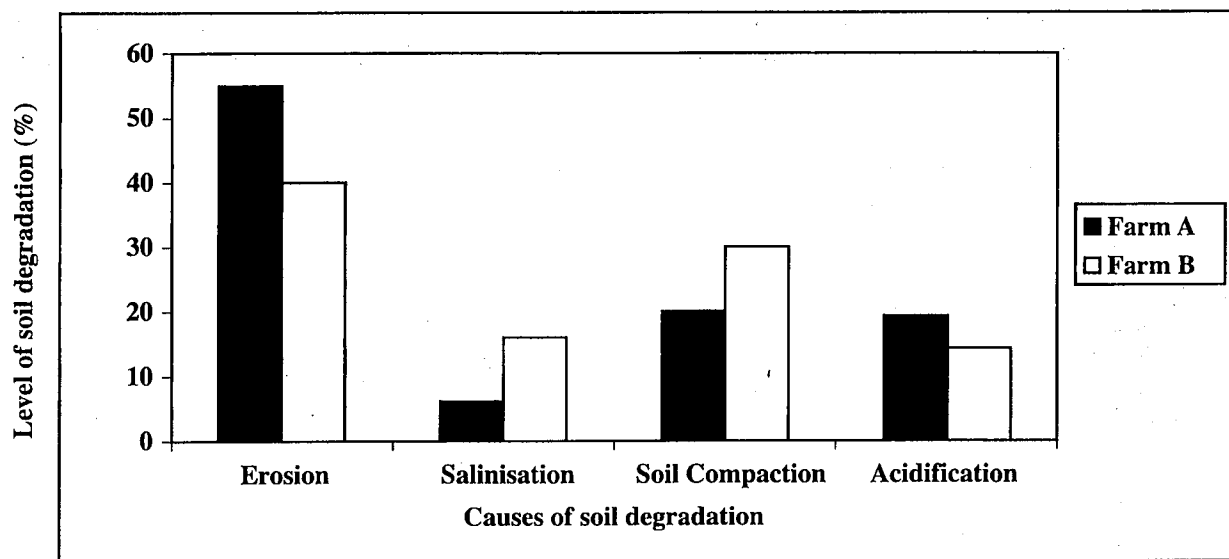


Figure 3. Causes of soil degradation

- (i) Comment on the causes of soil degradation as shown in Figure 3. Include specific values in your response. **[3 marks]**
- (ii) What mechanism caused the LEAST soil degradation? **[1 mark]**
- (iii) What percentage of soil degradation on Farm B was caused by erosion? **[1 mark]**
- (b) Conservation tillage would be a good measure for reducing soil degradation. Give TWO reasons to support the statement above. **[4 marks]**
- (c) Farmers in some rural areas are unlikely to consider increasing the level of mechanization in their operation. They have indicated that they have a number of concerns that need to be addressed before they can do so.
- (i) What is meant by 'mechanization in agriculture'? **[3 marks]**
- (ii) Suggest FOUR reasons why farmers in some rural areas may not wish to increase the level of mechanization in their farming operations. **[8 marks]**

Total 20 marks

MODULE 2

Answer ANY TWO questions.

4. (a) Study Figure 4, which shows the total sales of incandescent and fluorescent light bulbs in the Caribbean between 1996 and 2006, and answer the questions that follow.

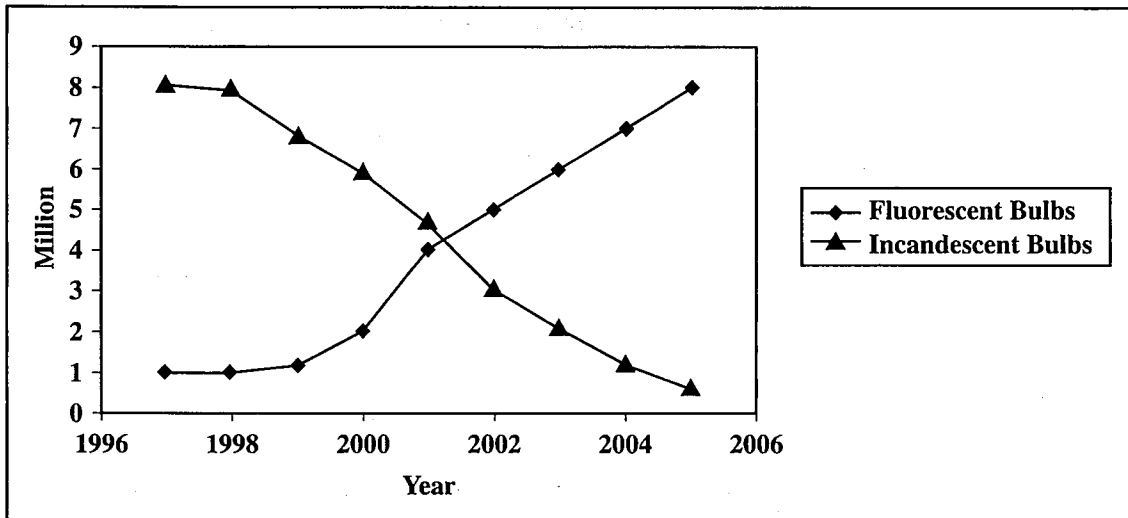


Figure 4. Total sales of incandescent and fluorescent light bulbs in the Caribbean

- (i) Describe the trend observed in sales of light bulbs in the Caribbean over the nine-year period shown in Figure 4. Include specific values in your response. [3 marks]
- (ii) In what year were there sales of 5 million fluorescent bulbs? [1 mark]
- (iii) How many bulbs were sold when sales for BOTH types of bulbs were equal? [1 mark]
- (iv) Suggest TWO reasons for the trend observed in the graph in Figure 4. [4 marks]
- (b) The use of 'combined cycles' is one of the fastest growing processes that is promoted to increase energy conservation and efficiency.
- (i) What is meant by 'combined cycles'? [3 marks]
- (ii) Would you recommend the use of combined cycles as a process for generating energy in your country? Suggest FOUR reasons to justify your response. [8 marks]

Total 20 marks

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5. (a) Table 1 shows trends in energy use by source and the willingness of people to convert to the use of a particular energy source in a Caribbean country.

TABLE 1: TRENDS IN ENERGY USE BY SOURCE

Energy Source	Annual Rate of Growth of Use (%)	Persons Willing to Use this Form of Energy (%)
Wind power	5	7
Natural gas	15	35
Solar	30	40
Geothermal	1	2
Hydropower	2	4

- (i) Outline the trend in annual rate of growth in use for the energy sources shown in Table 1. Include specific values in your response. **[3 marks]**
- (ii) Suggest ONE reason for the trends observed in (i) above. **[3 marks]**
- (b) The manager of a Power and Light company in a Caribbean country recently announced ~~plans to set up a conventional electricity generating facility within the coastal area. This facility, unlike the previous one set up by the company, will be located in close proximity to a coastal mangrove swamp.~~
- (i) Justify the locating of this conventional electricity generating facility within the coastal area as proposed by the manager. Give at least FOUR reasons to support your answer. **[8 marks]**
- (ii) Outline TWO environmental impacts that the manager of the company should consider before implementing his decision. **[6 marks]**

Total 20 marks

6. Study Figure 5, which shows the past and projected energy demand patterns from fossil fuels for three Caribbean countries, Country P, Country Q and Country R, and answer the questions that follow.

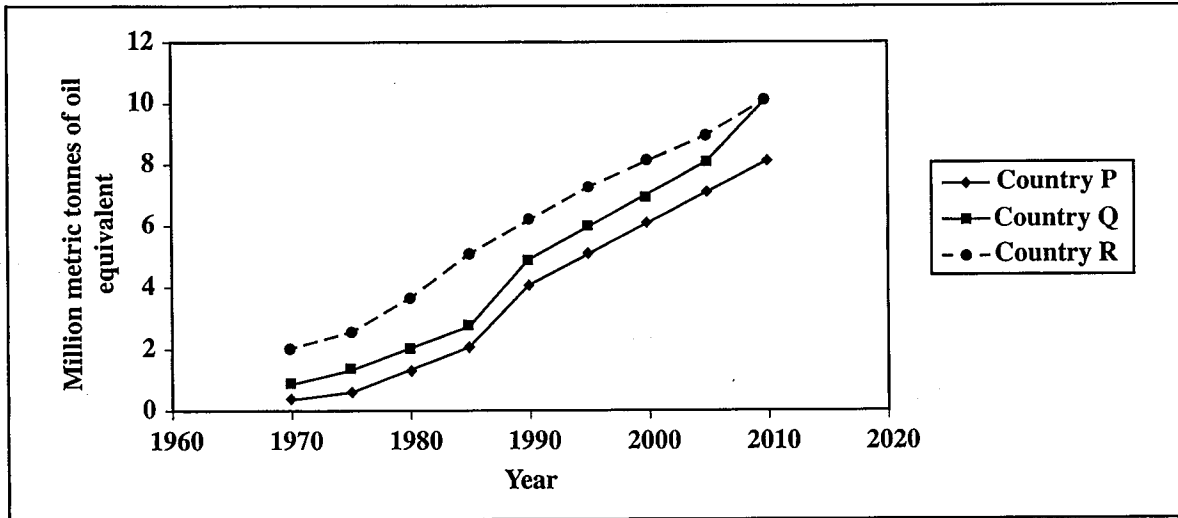


Figure 5. Past and projected energy demand patterns

- (a) Outline the overall demand pattern for energy from fossil fuels shown in Figure 5. Include specific values in your answer. [4 marks]
- (b) Suggest THREE reasons for the demand pattern shown in Figure 5. [6 marks]
- (c) The Minister responsible for the Environment in Country Q has cautioned citizens about its demand pattern as shown in Figure 5.
- Identify and discuss TWO environmental concerns that the Minister may have regarding the energy demand pattern as shown in Figure 5. [6 marks]
- (d) For EACH of the concerns identified in (c) above, make ONE recommendation. [4 marks]

Total 20 marks

MODULE 3

Answer ANY TWO questions.

7. (a) (i) List TWO primary air pollutants. [2 marks]
- (ii) Why is nitrogen dioxide considered a secondary air pollutant? [2 marks]
- (b) Ozone may be called a gas with a multiple personality, because sometimes it is considered a pollutant, and sometimes it is not.
- (i) Under what condition is ozone NOT considered a pollutant? [1 mark]
- (ii) Explain your response to (b) (i) above. [2 marks]
- (c) (i) Outline the formation of acid rain. [4 marks]
- (ii) Explain how acid rain results in the destruction of vegetation. [4 marks]

- (d) Figure 6 shows the results of an air quality monitoring exercise that was conducted to determine the impact of the construction of a new building on the airborne dust levels in a town. Use the data in the figure to answer the questions that follow.

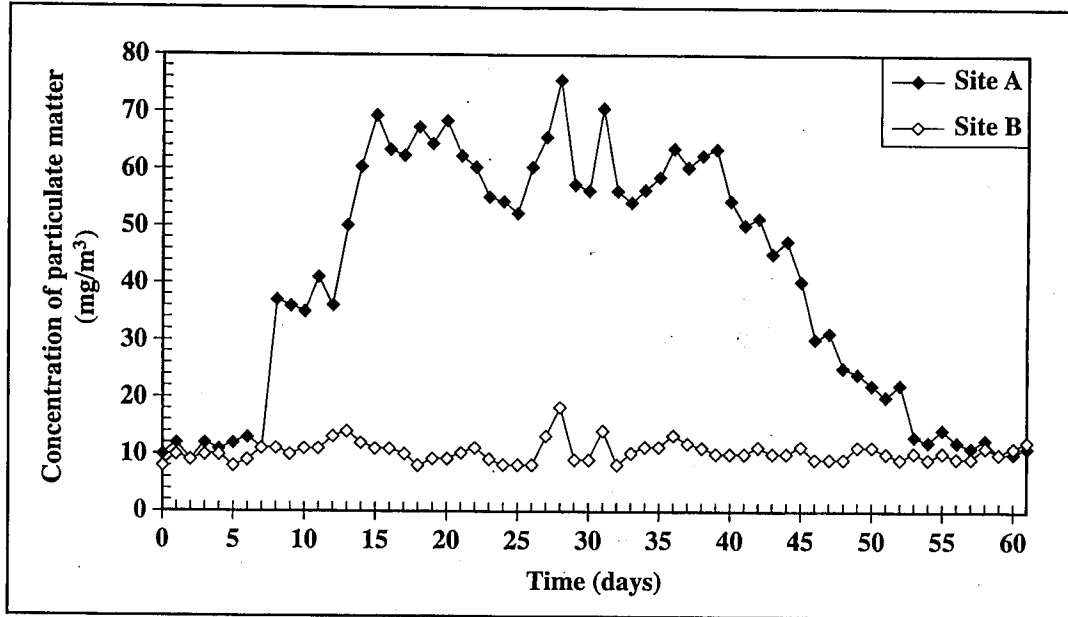


Figure 6. Results of an air quality monitoring exercise at two locations over two months

- (i) Identify which of Sites A and B was the experimental site and which was the control site. [1 mark]
- (ii) On which day did construction begin on the site? [1 mark]
- (iii) On which day did construction finally end on the site? [1 mark]
- (iv) Explain how you arrived at the answers to (d) (i) and (ii) above. [2 marks]

Total 20 marks

8. (a) Figure 7 illustrates the results of a monitoring exercise of a river that receives effluent from a sewage treatment plant. The dissolved oxygen levels in the river were monitored at intervals downstream of the plant, and the recovery of the oxygen in the river was determined. Study Figure 7, and answer the questions that follow.

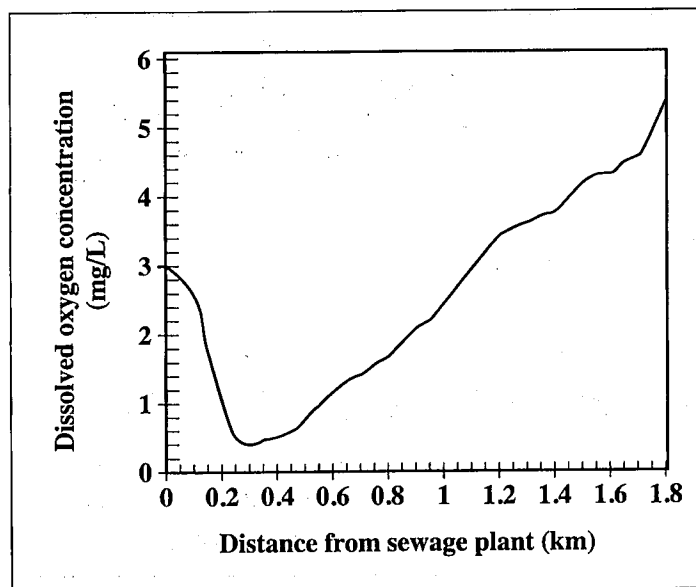


Figure 7. Concentration of dissolved oxygen downstream of a sewage treatment plant

- (i) At what distance from the sewage plant is the dissolved oxygen concentration at its LOWEST? [1 mark]
- (ii) What is the LOWEST dissolved oxygen concentration in the river? [1 mark]
- (iii) Given that fish must have a minimum of 3 mg/L dissolved oxygen for survival, for what distance downstream of the sewage treatment plant is the river completely without fish? [1 mark]

- (b) (i) Copy and complete Table 2 to show the name and ONE source of THREE different water pollutants EXCEPT nitrates. [3 marks]

TABLE 2: SOURCE OF WATER POLLUTANT

Name of Water Pollutant	Source
Nitrates	Nitrogen fertilizers, manure

- (ii) Outline how TWO of the pollutants you mentioned in (b) (i) above impact the environment. [4 marks]
- (c) It is suggested that pregnant women do not eat shark, due to the fact that shark meat may have high concentrations of mercury. Explain how a shark can exhibit possibly dangerous levels of mercury, if it does not come into contact with actual metallic mercury, and the concentration of mercury in the ocean is extremely low. [5 marks]

- (d) Table 3 provides information on three different methods for determining nitrates in water.

TABLE 3: EVALUATION OF METHODS TO DETERMINE NITRATES IN WATER

Parameter	Ion Selective Electrode	Spectrophotometric Method	Ion Chromatography
Detection limit (smallest concentration that the method can detect)	Low	Medium	Very low
Speed of analysis	Fast	Slow	Very fast
Cost per analysis	Medium	Low	Very high
Ease of use	Easy	Easy	Medium
Waste production per analysis	High	High	Low
Toxicity of waste	High	Medium	Low

- (i) You have been asked to recommend a nitrate-measuring method for your new laboratory. Assuming that you only need to consider the data provided in Table 3, which method would you recommend and why? [4 marks]
- (ii) Before you can make a final decision on which method to use for your laboratory, what is the MOST important factor that you must take into account? [1 mark]

Total 20 marks

9. (a) A bioremediation study was performed to determine how well a new bacterial culture would bioremediate oil-contaminated soil. The soil sample was tested before the experiment began and the initial soil concentration of total petroleum hydrocarbons (TPH) was found to be 540 mg/kg. Study Figure 8, which illustrates the results of this study, and answer the questions that follow.

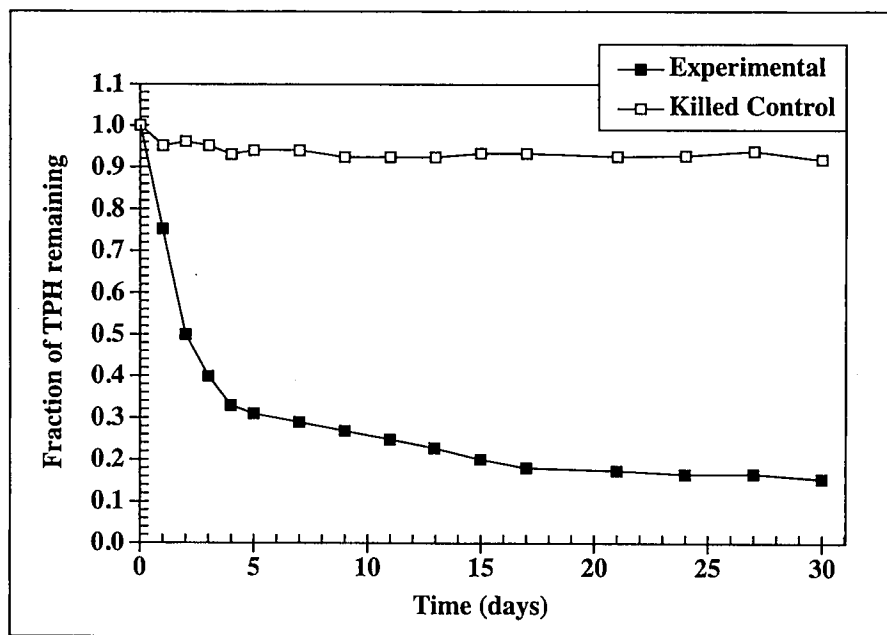


Figure 8. Results of a bioremediation study for an oil-contaminated soil

- (i) Define the term 'bioremediation'. [2 marks]
- (ii) How long did it take for the TPH level to drop to 60% of its initial level? [1 mark]
- (iii) What was the loss in actual TPH concentration by the end of the study? [2 marks]
- (iv) How do you know that it was the new bacterial culture that accomplished the reduction indicated in Figure 8? [3 marks]
- (b) (i) Outline the functioning of incineration as a solid waste disposal technique. [2 marks]
- (ii) Incineration is being considered in a number of Caribbean islands as an alternative to open landfills as a waste disposal technique. Discuss this proposal. Include TWO advantages and THREE disadvantages of incineration in your discussion. [10 marks]

Total 20 marks